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**Published ahead of Print**

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Accepted for Publication: 29 March 2025

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Medicine & Science in Sports & Exercise, Publish Ahead of Print
DOI: 10.1249/MSS.0000000000003726

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**Conflict of Interest and Funding Source:** This study was internally funded and received no external grant(s) or other funds. The authors declare no conflicts of interest.

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**ABSTRACT**

**Purpose:** Studies on heat acclimation (HA) involving high-intensity exercise report impairments in time trial (TT) performance in the heat immediately after HA. This study aimed to determine whether a high-carbohydrate (HC) diet can enhance 3.22 km run TT performance in the heat following exercise-intensive short-term HA. **Methods:** Fourteen healthy military-age males were assigned to either a HC diet (70 %kcal carbohydrate (CHO), n = 7) or lower CHO comparison (COMP, 35 %kcal CHO, n = 7) to be consumed during and for one week after a 6-day HA protocol. **Results:** Baseline descriptive statistics, heat tolerance, and run performance were similar between groups ($p > 0.05$). Both groups acclimated, showing reduced heart rate, rectal and skin temperature, and increased sweat rate (each $p \leq 0.03$) during heat stress testing one day and five days after HA. The HC group demonstrated a faster TT (959 ± 103 sec vs. 1067 ± 172, $p = 0.02$, $g = 0.71$) than COMP one day post-HA. HC showed improved TT performance from baseline at one day ($p = 0.01$, $g = 0.59$) and five days post-HA ($p = 0.04$, $g = 0.59$). The HC group showed greater TT improvement at one day (-7.0 ± 4.9 % vs. -0.7 ± 4.7 %, $p = 0.03$, $g = 1.23$) and five days post-HA (-8.4 ± 4.9 % vs. -2.8 ± 3.9 %, $p = 0.01$, $g = 1.18$), with no group differences in HA outcomes or relative strain during TT. **Conclusions:** Insufficient CHO intake during heavy daily exertion in unacclimated heat stress, such as in short-term HA, may affect post-HA performance outcomes unless addressed with additional CHO consumption.

**Key Words:** FATIGUE, NUTRITION, READINESS, WARFIGHTER

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## INTRODUCTION

Military operations can occur in any climate, including hot environments, threatening the health and readiness of warfighters. Heat acclimation (HA) protocols employed before or upon arrival can reduce these risks, but traditional protocols may exceed two weeks and contribute to detraining in trained individuals, as low-intensity exercise is often used for HA. (1) Short-term heat acclimation (STHA) uses intensive exercise or increased heat for 5-7 days, providing thermoregulatory and circulatory benefits. (2) However, studies that have conducted intense exercise during STHA have reported diminished performance in the heat after STHA. (3, 4) Such performance degradation will impair mission readiness and threaten warfighter effectiveness. Warfighters frequently leverage high-intensity exercise in their physical training to maximize results given limited available time. Adding heat stress to their training without recovery countermeasures, such as nutritional intervention, may lead to undesirable losses in performance readiness. Daily perceived exertion and fatigue, combined with speed and acceleration data, indicate that consecutive days of intense training result in non-linear performance decrements until the stress is mitigated or removed. (5, 6)

Heat strain may impair performance during consecutive exposures, partly due to accelerated depletion of muscle glycogen at both moderate and high work intensities (7), along with negative neural feedback (8–10), which ultimately reduces high-intensity physical performance potential. Unaccustomed heat stress raises the oxidation rate of glycogen reserves (7, 11–15) and increases epinephrine concentrations. (13, 16, 17) While successful heat acclimation returns the oxidation rate of endogenous carbohydrate (CHO) and epinephrine levels in hot conditions to those comparable to temperate conditions (18–21), the timeline of these adaptations

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remains unclear. Sports athletes often undergo HA in a mostly predictable life situation with ready access to food and reprieve from environmental stress in their living quarters. However, warfighters executing an exercise-based HA protocol in a field environment may frequently encounter other environmental stressors, face inconsistent food availability, and deal with additional physical demands related to mission execution that affect glycogen reserves. A sustained and progressive decline in muscle glycogen during HA can significantly affect mission readiness unless addressed in the diet.

A diet high in CHO is an effective strategy for sustaining physical performance in tactical athletes. (22–25) Additional dietary CHO is advised during moderate-to-high intensity exercise in hot conditions (26) to counteract increased CHO oxidation rates and promote positive neurological effects to sustain performance. (27–30) Acute supplementation of CHO is a recognized ergogenic aid for enhancing physical output in events that deplete glycogen reserves through high activity intensity and/or duration. (23, 24, 27, 31, 32) There are general recommendations to increase CHO intake when exercising under heat stress (26); however, few studies evaluated CHO provision as a component of HA outcomes. Additionally, a recent review addressed the nutrition gap in HA literature and examined other nutrition-related physiological variables that may affect physical capacity in the heat. (33)

The objective of this study was to assess the influence of a eucaloric, isonitrogenous high-CHO diet on running performance in the heat immediately after HA and after several days of recovery compared to a pair-matched lower-CHO comparison group. We hypothesized that both groups would successfully heat acclimate, but only the increased CHO provision would enhance

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performance in the heat immediately following STHA compared to a diet lower in CHO. Both groups would improve performance over baseline after a multi-day recovery period.

## METHODS

### Participants

The study received approval from the university Institutional Review Board (#00006938) in accordance with the Declaration of Helsinki. Healthy males of military age (18 to 40 years) were recruited from the local population. After the study's risks and demands were clearly communicated to the participants, written informed consent was obtained. Participants underwent screening and were required to be free of any existing musculoskeletal injuries, cardiometabolic or respiratory diseases, prior heat-related illnesses, and were not prescribed medications that could affect thermoregulation or exercise tolerance. The study team conducted multi-pass interviews to quantify weekly training volume. If participants exercised at a moderate-to-high intensity for at least 250 minutes per week, they were eligible to participate in the study. All heat-related study visits took place in the northeast United States between October and April, when the mean ambient temperature was 5.3 ºC. (34) No participants reported exercise heat exposure in the four weeks prior to enrollment and reported not being acclimated to heat. Refer to Figure (1) to review the overall study flow.

### $VO_{2peak}$

Aerobic fitness was assessed using a modified Astrand treadmill protocol (35) followed by a verification test (36) in a hot room (37.1ºC, 38.4 % relative humidity (RH)) before and after HA. Participants were instructed to refrain from alcohol, caffeine, and exercise for at least 12 hours

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before testing, be fasted for at least two hours, and arrive to the lab euhydrated defined as a urine-specific gravity < 1.020 (Master-URC, ATAGO Co., Tokyo Japan). (37) Participants recorded a nude body mass (Seca 217, Seca North America, CA, USA), self-inserted a rectal thermistor (400 series, Medline Inc., IL, USA) 10 cm past the anal sphincter to monitor rectal temperature ($T_{re}$), then thermocrons (DS1822, Maxim Integrated, CA, USA) were affixed to the right upper chest, triceps, quadriceps, and calf to assess mean skin temperature ($T_{sk}$). (38) Respiratory gases were recorded using an indirect calorimeter (TrueOne 2400; ParvoMedics, UT, USA) located outside the chamber with continuous HR monitoring (T31, Polar, NY, USA). Blood lactate was assessed (Lactate Plus, Nova Biomedical, MA, USA) before and immediately after completion of both the ramped and verification trial. Ramp testing concluded upon volitional exhaustion or at the participants request, after which participants rested in a cool room for 20 mins before re-entering the chamber to complete the verification trial. The fixed-speed two-stage verification involved two mins at 60% $W_{max}$ achieved in the ramp test, followed by time to exhaustion at 105% $W_{max}$. Participants were eligible to participate in the remainder of the study if the final $VO_{2peak}$ values attained in both tests were between 45 to 60 $ml \cdot kg^{-1} \cdot min^{-1}$, a representative range observed in multiple studies of male military personnel. (39–41)

**Resting Metabolic Rate (RMR)**

A metabolic assessment determined appropriate caloric intake to maintain body mass consistency between pre- and post-HA testing and to set carbohydrate targets based on energy percentage consumed. Participants arrived at the lab within one hour of waking and fasted from all non-water food and beverages for 12 hours. Participants then laid supine on an exam table in a quiet, dark room for at least 20 mins, after which an RMR canopy was placed over the participant's

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head, assessing respired gases via indirect calorimeter (TrueOne 2400; ParvoMedics, UT, USA) for another 30 mins in accordance with manufacturer directions. Once RMR was determined, a single trained investigator applied a physical activity adjustment to multiply RMR by 1.6 to 1.9 depending on lifestyle energy demands, a range reflective of active individuals exercising daily. (42) Participants were pair-matched by aerobic fitness and weekly training volume, then randomized to one of two diet conditions: high-CHO (HC) or a lower-CHO comparison (COMP) diet.

**Dietary Prescription**

Table 1 outlines the dietary targets for both the HC and COMP groups before baseline testing and during the HA and post-HA intervention periods. The HC group's target was to consume 70% of calorie needs from CHO, which translated to approximately 6 $g \cdot kg^{-1} \cdot d^{-1}$ depending on body mass and RMR, an intake level recommended for tactical athletes (23, 31) and for exercising under heat stress (26). The COMP group's target for CHO was 35% of calorie needs, equating to roughly 3 $g \cdot kg^{-1} \cdot d^{-1}$, the lowest recommended CHO level for active individuals. (23) The COMP prescription aimed to create a stark contrast to HC and simulate suboptimal nutrition practices in a warfighter context, where carbohydrate-rich military rations may not be fully consumed, and high-fat snacks are used to meet caloric needs. Both HC and COMP were instructed to consume 1.4 $g \cdot kg^{-1} \cdot d^{-1}$ protein, a moderate intake level recommended for tactical athletes (31) training 1-2 hours per day. After meeting CHO and protein targets, fat intake was prescribed to fill the remaining calorie needs.

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Participants were instructed on using food weight scales (Arc Pro, Salter, Manchester, UK) to assess dietary intake accurately. At the RMR visit, they submitted a 24-hour food record, and an investigator conducted a multi-pass assessment to determine typical meals and patterns. (32) These foods and patterns served as a basis for nutrition counseling by a Registered Dietitian to aid participants in meeting their prescribed dietary criteria. Participants were counseled on a temporary diet to consume 50% of their needs from CHO, reflecting the approximate mean intake for young active males (43), for two days before the initial heat stress test to standardize pre-test CHO availability. Twenty-four hours before the first HA session (44), participants began following their assigned diet protocol (HC or COMP) and recorded their intake, which was used for a multi-pass assessment by a single trained Registered Dietitian.

**Pre-Acclimation Heat Stress Test (HST1)**

A standardized protocol was used to (a.) determine the physiological response to a specific exercise and heat load and (b.) assess subsequent aerobic performance under heat stress. Participants followed the same pre-trial and setup procedures as the $VO_{2peak}$ assessment with the addition of a 50 %kcal CHO diet for 48 hours prior to the visit as in Table (1) and 24-hour food record. Participants were instructed to consume a consistent self-selected meal containing approximately 100g CHO 3-4 hours before each trial, then to arrive fasted for at least two hours, and to wear the same shirt, shorts, and shoe ensemble for each HST. Resting HR, $T_{re}$, $T_{sk}$, and blood lactate were recorded before entering a hot environmental chamber (37.2 ºC, 41.7 % RH). Once in the chamber, participants performed a modified U.S. Army ATP 7-22 preparatory drill calisthenics warm-up (45) and then began a 45 min walk at 1.56 $m \cdot s^{-1}$ on a motorized treadmill (Pro XL, Woodway, WI, USA) at an incline eliciting 50% $VO_{2peak}$ (20) to determine heat tolerance.

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Treadmill settings remained consistent for all HSTs. Heart rate, $T_{re}$, and $T_{sk}$ were recorded every five mins, and respiratory gases were collected at mins 5, 10, 25, 30, and 45 by an indirect calorimeter located outside the chamber. Rates of carbohydrate ($CHO_{ox}$) and fat oxidation ($FAT_{ox}$) were assessed at minutes 10, 30, and 45 then averaged across the walking effort. (46) Tepid water was provided *ad libitum* throughout and recorded. Perceived fatigue (0-10 scale, 0 = no fatigue, 10 = complete fatigue)(47) and blood lactate were assessed at the conclusion of the walk, then participants were given five mins to rest in the chamber before the time trial (TT).

Aerobic performance in the heat was assessed on a motorized treadmill through a self-paced 3.22 km run, the distance used for U.S. Army fitness assessments. (48) Participants were blinded to the distance covered, speed, and time elapsed. The starting belt speed was set to a familiar but blinded jogging pace (2.2 $m \cdot s^{-1}$, same as $VO_{2peak}$ test) with the incline set to 1% grade; then participants were given control of the treadmill speed, at which point distance and timing began. Verbal updates on the distance covered were provided every 0.8 km. A single investigator led each performance test to ensure consistent vigorous motivation was given to each participant. Throughout the run, HR, $T_{re}$, $T_{sk}$, speed, and elapsed time were recorded every 0.4 km. Mean percentage of theoretical $HR_{max}$ (208 – 0.7*age) (49) was calculated across each running timepoint to quantify effort. After completing the TT, participants were removed from the environmental chamber and recorded a towel-dried nude body weight to assess sweat losses after accounting for fluid intake and urination.

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**Heat acclimation week**

Participants completed six exercise HA sessions within a seven-day period while refraining from any other exercise during HA. Of the six HA visits, at least three sessions (namely days 1, 4, and 6) were conducted on a treadmill (Curve Trainer, Woodway, WI, USA), while the remaining visits included either treadmill sessions or low-impact sessions on a stationary bike (BikeERG, Concept2, VT, USA) or rowing ergometer (RowERG, Concept2, VT, USA). The exercise modality was participant-driven based on individual comfort and familiarity, although session intensity goals remained consistent. Each session began with an assessment of USG to ensure euhydration, then self-placement of the rectal thermistor and HR strap, and calisthenic warm-up as in HST1. After entering a hot room (39.3 ºC, 38.3 % RH), participants performed a high-intensity exercise protocol until $T_{re}$ reached 38.5 ºC. Treadmill sessions included repeated two-min run intervals at 105-115% of HST1 run pace with 90 sec walking recovery, while bike and rowing workouts involved two-minute intense efforts at RPE 15-18 followed by 90 sec low-intensity recovery. (50) Once participants reached a $T_{re}$ of 38.5 ºC, they stayed on their chosen machine and walked, jogged, cycled, or rowed at a low- to moderate intensity as needed to maintain $T_{re}$ 38.5 to 39.0 ºC for at least 60 minutes. The distance covered during the three mandatory treadmill workouts was used to quantify average workload during HA. Heart rate and $T_{re}$ were recorded throughout, along with total heat exposure and time spent at or above 38.5 ºC $T_{re}$. Participants were provided 6 % CHO electrolyte solution and tepid water *ad libitum*, noting that the CHO beverage consumed counted toward their daily nutrient targets, as annotated in the Supplemental Table 1 (Supplemental Digital Content).

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## Post-HA Testing

Two post-HA HSTs were conducted: following one day of rest from the final HA session (HST2) and five days after HA (HST3) to assess HA retention. A second $VO_{2peak}$ and verification trial in the heat was performed between HST2 and HST3 to assess for any changes in fitness. Decay of heat adaptations were expected to be minimal in this post-test period. (51) The final trials were scheduled at the same time of day with similar pre-trial self-selected meals and mimicked procedures used in the first respective HST and $VO_{2peak}$ visits. Participants maintained their assigned diet throughout and refrained from any additional moderate-to-intense exercise during the post-testing week. Dietary intake was recorded in the 24-hour period preceding HSTs.

## Statistical analysis

Sample size estimation (G*Power, Heinrich-Heine-Universität Düsseldorf, Germany) was based on normalized TT improvement data from both STHA (-59 ± 34 s) (19) and high-CHO (-24 ± 12 s) (52) interventions on running performance. The initial sample model was based on a mixed model repeated measures analysis of variance (ANOVA) recommending seven participants per group. By study completion, the statistical approach evolved to include a mixed-model analysis of covariance (ANCOVA) with HST1 as a covariate and Bonferroni corrections for post-hoc pairwise comparisons (53) to evaluate group TT performance and any difference in HA outcomes (alpha = 0.05). A mixed-model repeated measures ANOVA was employed to compare group macronutrient intake and detect the presence of HA adaptations within groups, with Tukey's HSD for post-hoc pairwise comparisons. Heat stress and running performance outcomes were analyzed for normality using a Shapiro-Wilk test. Dietary intake was analyzed using Nutritionist Pro v. 7.9 (Axxya Systems, WA, USA). All statistical analyses were performed using SPSS v. 29.0.2 (IBM, NY,

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USA) and Prism v.10 (GraphPad, CA, USA). Data are presented as mean ± standard deviation with p-values, partial eta squared ($\eta^2_p$), ANCOVA time and group effects, and Hedges g effect sizes are presented for significant differences unless otherwise noted.

## RESULTS

### Pre-HA comparisons

Group baseline descriptive variables are listed in Table 2; there were no significant differences in age, weight, total body surface area (54), pre-HA $VO_{2peak}$, or habitual moderate-to-intense training volume per week (all $p \geq 0.05$). Table 3 overviews nutritional intake and results of the walking portion of the HSTs. At HST1 there were no differences between groups in nutritional intake preceding the trial. There were no between-group differences in 3.22 km TT performance at HST1 (Table 4, $p > 0.60$), with both groups exhibiting a high degree of cardiovascular strain (final HR and average $\%HR_{max}$) and considerable hyperthermia at the conclusion of the run.

### Acclimation week

There was a significant group difference in CHO consumption during HA ($p < 0.01$). CHO intake was greater in HC vs. COMP at HA days 1 and 4 (Figure 2; HC: 67.4 ± 4.8 %kcal and 69.5 ± 3.4, respectively vs. COMP: 32.5 ± 1.8 %kcal and 36.4 ± 2.9, $p < 0.01$, g = 8.95 and 9.74). Additionally, there was a group difference in fat intake ($p < 0.01$) with lower intake in HC (15.0 ± 3.5 %kcal and 13.0 ± 2.6, respectively vs. COMP: 49.2 ± 3.6 %kcal and 45.3 ± 3.4, $p < 0.01$, g = 8.96 and 9.93). There were no group effects for relative energy intake ($p > 0.56$) nor effects for protein consumption ($p > 0.34$) (Table 3). There were no group differences in total or high-intensity

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training volume (Table 2). Both groups spent a comparable amount of time with $T_{re} \geq 38.5$ ºC, exceeding 60 mins per day, with a similar average daily heat exposure of approximately 90 mins.

**Post-HA Testing**

Table 3 summarizes nutritional intake and walking heat stress outcomes at the post-HA HSTs. There was no effect of time on body weight ($p > 0.51$), as both groups maintained their body mass between HST1 and HST3. They sustained a significant difference in CHO intake ($p < 0.01$) and fat intake ($p < 0.01$) at both HST2 and HST3, with no group differences in relative protein or energy consumption (all $p > 0.2$). Controlling for HST1 responses, the groups showed a time effect ($p < 0.01$) at HST2 for end-walk HR ($\eta^2_p = 0.76$), $T_{re}$ ($\eta^2_p = 0.62$), $T_{sk}$ ($\eta^2_p = 0.47$), and sweat rate ($\eta^2_p = 0.91$) immediately following HA. At HST3 following five days of recovery, the time effects remained ($p < 0.01$) for HR ($\eta^2_p = 0.71$), $T_{re}$ ($\eta^2_p = 0.82$), $T_{sk}$ ($\eta^2_p = 0.50$), and sweat rate ($\eta^2_p = 0.80$). There were no between-group differences in HA adaptations at HST2 or HST3.

Metabolic adjustments were identified after HA, with time and group effects found for both $CHO_{ox}$ and $FAT_{ox}$ after accounting for HST1 responses (Figure 3). At HST2, there was a time effect for $CHO_{ox}$ ($p < 0.01$, $\eta^2_p = 0.64$) along with group difference ($p = 0.05$, $\eta^2_p = 0.30$), and while the time effect continued at HST3 ($p < 0.01$, $\eta^2_p = 0.55$), a group effect was not found ($p = 0.24$). Within the HC group, mean $CHO_{ox}$ rates increased compared to HST1 at both HST2 ($p = 0.04$, $g = 0.44$) and HST3 ($p = 0.03$, $g = 0.51$); COMP showed no changes in $CHO_{ox}$ at either HST2 ($p = 0.14$) or HST3 ($p = 0.83$). At HST2, there was a time ($p = 0.05$, $\eta^2_p = 0.31$) and group effect for $FAT_{ox}$ ($p = 0.03$, $\eta^2_p = 0.35$); the HC group exhibited suppressed $FAT_{ox}$ at HST2 ($p < 0.01$, g

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= 1.24) whereas COMP saw no significant changes in $FAT_{ox}$ ($p = 0.49$). By HST3 there was no time ($p = 0.44$) or group effect ($p = 0.58$) in $FAT_{ox}$.

After controlling for HST1 run performance, there was a significant time effect ($p < 0.01$, $\eta^2_p = 0.90$) and group difference in run time ($p = 0.02$, $\eta^2_p = 0.40$) at HST2 with the HC group improving 3.22 km TT performance from baseline ($p = 0.01$, $g = 0.53$) whereas COMP did not improve ($p = 0.80$)(Figure 4). By HST3, the time effect for TT performance remained ($p < 0.01$, $\eta^2_p = 0.82$), but there was no group effect ($p = 0.07$). The HC group again exhibited a faster TT than at their HST1 ($p = 0.03$, $g = 0.59$), but COMP did not ($p = 0.11$). The group differences at HST2 appear driven by reductions in the time to complete the second half of the TT compared to HST1 (Table 4), with time effects ($p < 0.01$, $\eta^2_p = 0.75$) and group effects present at HST2 ($p = 0.03$, $\eta^2_p = 0.35$). There were no group differences in time to complete the first half of the run at HST2 ($p = 0.09$) or HST3 ($p = 0.36$) despite time effects (HST2: $p < 0.01$, $\eta^2_p = 0.81$, HST3: $p < 0.01$, $\eta^2_p = 0.72$). The mean improvement in trial time (seconds) was greater for HC at HST2 compared to COMP ($p = 0.04$, $g = 1.13$), but not at HST3 ($p = 0.23$). As a percentage of TT improvement, HC experienced a 7.0 ± 4.9 % reduction in TT duration at HST2 compared to 0.7 ± 4.7 % reduction in COMP ($p = 0.03$, $g = 1.23$) and 8.4 ± 4.9 % vs. 2.8 ± 3.9 % improvement at HST3 between HC and COMP, respectively ($p = 0.01$, $g = 1.18$). There were no within- or between-group differences in end-run HR, $T_{re}$, mean $\%HR_{max}$ or final RPE in any TT (all $p > 0.05$).

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## DISCUSSION

Our key findings include (a.) the HC group showed improvements in 3.22 km TT in the heat at HST2 one day after HA and at HST3 five days after HA, (b.) the COMP group did not show improvement in either post-HA TT, and (c.) after controlling for baseline run time, the HC group ran faster than COMP at HST2 but not HST3. The improvements in running time in the HC group appear driven by an average reduction of 9% in time to complete the second half of the TT. By the latter half of the TT, participants had been exercising under heat stress for approximately 60 to 70 mins, the time frame at which additional CHO is recommended during exercise. (23, 32) These data potentially underscore the ergogenic importance of dietary intake, particularly carbohydrates, for achieving anticipated performance enhancements in the heat after a HA protocol that results in high internal and external load.

The HC group achieved the expected level of performance improvement seen in previous studies, despite being in a pre-fatigued state from the 45-min walking effort immediately before the TT. However, the lack of significant TT change in the COMP group seems to be notably linked to the reduced CHO intake. During HA, both HC and COMP spent a similar amount of time above 38.5ºC $T_{re}$ with comparable training volumes. Both exhibited significant physiological adaptation at HST2 and HST3 (lower $T_{re}$, $T_{sk}$, HR, increased sweat rate) and experienced a severe but comparable degree of global strain during the TTs, indicating that nutritional differences could be the main factors affecting performance.

One HC participant showed notable improvement in run times from HST1 to HST2/HST3 (1253 s to 1099 s and 1008 s), despite no changes in HR or hyperthermia during TTs. Re-analyzing

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without this participant's data did not alter the HC group's significant difference from HST1 times at HST2 ($p = 0.02$) and HST3 ($p = 0.02$). The HC group's improvement in heated TT performance after HA was similar to the 8% from Lorenzo et al. (55) and the 7 ± 7 % expected change found by Tyler et al. (56) Many studies influencing the 7 ± 7 % improvement are based on steady-state exercise HA, possibly less fatiguing than our study's workload. Additionally, Tyler et al.'s cited studies often utilized separate TTs or portions of TTs to assess HA outcomes, while ours integrated the walking HST and TT in one visit. This method enhanced ecological validity for combat operations and purposefully aimed to pre-fatigue participants with moderate-intensity walking, revealing performance differences in the TT tied to dietary factors.

Our study complements Bennett et al. (57) in several ways. They found that, compared to a matched two-week temperate training with similar daily CHO intake (6 $g \cdot kg^{-1} \cdot d^{-1}$) and timing (train low CHO), heat training impaired cycling performance immediately after HA despite adequate adaptations, while the temperate group improved performance. They also noted that the heated group showed no expected increase in $FAT_{ox}$ during post-testing, suggesting that the heat-induced shift to $CHO_{ox}$, along with higher internal load, caused some overreaching. In our study, we employed approximately the same 6 $g \cdot kg^{-1} \cdot d^{-1}$ CHO prescription for the HC group during a shorter HA protocol and found suppressed $FAT_{ox}$ and higher $CHO_{ox}$ in post-testing; however, the HC group did show improved endurance performance. We did not control CHO timing during HA training or standardize pre-trial meals as Bennett et al. did, but both HC and COMP experienced similar conditions (heat exposure time, training volume, time $T_{re} > 38.5$ ºC), lending validity to our comparisons. Thus, both studies concluded that restricting CHO during intense heat training should be avoided if performance is critical.

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Research on carbohydrate (CHO) manipulation in endurance shows inconsistent results. Williams et al. found that during a 30 km run time trial, participants on a CHO-supplemented diet for seven days improved overall time by 3% and speed on the last 5 km by 6%, compared to a control group with added fat and protein. (52) Later, Pitsiladis et al. compared 10 km run time trials in trained runners on either 40 %kcal or a progressively increasing 55- 70 %kcal CHO diet over seven days, reporting no differences. (58) These findings might be affected by the trials' short duration and the participants' high training status. Our study included a diverse group with varying running abilities, similar to a military population while controlling for baseline responses and performance for reliable analysis.

Individuals participating in general fitness training for 45 to 75 mins per day are recommended to consume 3 to 5 $g \cdot kg^{-1} \cdot d^{-1}$ CHO, increasing to 5 to 8 $g \cdot kg^{-1} \cdot d^{-1}$ for those undergoing 2 to 3 hours of intense training most days of the week. (23) Relative CHO intake was compared at HST2 and HST3, with the COMP group averaging 3.2 $g \cdot kg^{-1} \cdot d^{-1}$ CHO while the HC group averaged 5.9 $g \cdot kg^{-1} \cdot d^{-1}$. During HA and at each HST, participants exercised at moderate-to-high intensities for 70 to 90 mins while under heat stress, positioning their CHO needs within a general fitness criterion, albeit at the lower end for COMP. Glycogen status in the HC group could have significantly increased, reaching 80-95% replenishment within 12-24 hours of training or testing based on previous research involving similar CHO volumes (44). In contrast, reduced CHO diets (e.g., 2-3 $g \cdot kg^{-1} \cdot d^{-1}$) have shown minimal to partial (5-30%) replenishment after exhaustive exercise. (59, 60)

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Our HA was intense but not exhaustive, and a rest day was scheduled before HST2. However, this might not have provided sufficient time for the COMP group to address potential declines in glycogen status throughout the HA week. Both groups showed no differences in protein consumption, maintaining body mass and energy balance relative to RMR and activity goals, indicating similar recovery stimulus aside from dietary CHO and fat contrast. It remains unclear if a high-CHO recovery period can improve performance after HA in a lower-CHO state, similar to military contexts where warfighters rotate out of high-demand operations and obtain better access to food and rest. One study found that consuming a high-CHO diet for three days after ten days of very-low CHO training improved performance vs. training with a habitual diet (61). Recovery interventions for suboptimal nutrition strategies in HA are an area for future research.

There are limitations and considerations to discuss, starting with the selected dietary parameters. The 35 %kcal CHO prescription in the COMP group was designed to provide the lowest acceptable amount of CHO for active individuals (e.g.; 3 $g \cdot kg^{-1} \cdot d^{-1}$)(23, 31); notably it is below the average intake for active young men, which is between 47.6 and 48.9 %kcal CHO. (43) Our study found a consistent difference in CHO and fat intake between groups. With the rigorous implementation of nutrition controls, future studies could utilize a comparison diet that more accurately reflects habitual intake to contrast with a high-CHO diet intervention. The increased fat intake in the COMP group was within acceptable criteria for athletes, and equivalent intakes with energy balance have been found to have no significant ill effects on exercise recovery. (62) The HC group's dietary fat reduction to < 20 %kcal should be short-term due to risks of essential fatty acid and vitamin deficiencies, which vary by individual. (32) This study's nutrition outcomes relied on self-reported and weighed dietary intake with multi-pass interviews by a Registered

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Dietitian, a method found to have the lowest variation from doubly-labeled water comparisons. (63) No self-reported dietary analysis is free from bias and under-reporting. Nonetheless, we believe our group comparisons are valid since the under-reporting likelihood was similar for both groups.

Other internal validity concerns included the inability to blind participants to their group assignment with natural human variability in perceived maximal effort during the running TT. We quantified TT effort through cardiovascular strain (HR, $\%HR_{max}$), perceptual strain (RPE, fatigue), and objective hyperthermia ($T_{re}$, $T_{sk}$), finding no difference between groups, though internal validity is a persistent issue with free-living human research. Additionally, the HA exercise program was neither pair-matched nor randomized, as it depended on participants' skill, capacity, and the work required to reach $T_{re}$ targets. As for ecological validity, our study ensured energy balance for valid comparisons. However, warfighters are susceptible to significant energy deficits (64) dependent on multiple external factors. Future studies should evaluate varying degrees of energy intake on HA recovery and include longer, more intensive exercise during heat stress tests to evaluate physiological adaptations differentiating high-CHO from others groups. Other considerations include implementing familiarization TTs, evaluating female participants, and expanding HA protocols with longer duration and/or without rest days to further stress recovery and glycogen reserves.

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## CONCLUSIONS

Recreationally trained males following a high-CHO diet (70 %kcal) throughout an exercise-intensive six-day heat acclimation protocol performed better than a lower-CHO comparison diet (35 %kcal) during a TT in the heat one day after acclimation but the group performance did not differ after a five-day recovery period. Improvements in run time were observed in the second half of both high-CHO post-acclimation run trials. Both diet groups sufficiently acclimated to heat stress with comparable heat acclimation volume (training, time, and thermal load), suggesting potential for a diet-driven improvement in endurance performance. During STHA utilizing moderate-to-high exercise intensity exercise with isothermic procedures, we recommend increasing CHO in the diet, or at least not restricting them below a typical diet to realize performance improvements.

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**Acknowledgements**

This study was internally funded and received no external grant(s) or other funds. We would like to express our gratitude to our outstanding and dedicated study participants who made this possible. The authors declare no conflicts of interest. The results are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The work presented here does not constitute endorsement by the American College of Sports Medicine.

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## FIGURE LEGENDS

**Figure 1**: Schematic of study flow. NOTE: CHO = Carbohydrate; HST = Heat stress test; RMR = Resting metabolic rate analysis. = 24-hour diet record analysis; = Running performance test in the heat; = thermal tolerance testing.

**Figure 2**: Nutritional comparisons across the study between high-carbohydrate (HC) intervention group and lower-carbohydrate comparison (COMP) at the baseline heat stress test (HST1), during heat acclimation (HA) days 1 (D1) and 4 (D4), and post-heat acclimation tests (HST2 and HST3). The dotted line represents specific targets for calories and protein; note there were not specific fat consumption targets as fat was intended to fill remaining calories after carbohydrate (70 vs. 35% kcal) and protein (1.4 $g \cdot kg^{-1}$) targets were considered. * = difference between groups, $p < 0.05$.

**Figure 3**: Carbohydrate and fat oxidation rates during the walking portion of each heat stress test between high-carbohydrate intervention group and lower-carbohydrate comparison (COMP). a = different from HST1, * = difference between groups, $p < 0.05$.

**Figure 4**: Comparison of 3.22 km running performance during the baseline heat stress test (HST1) and after following the assigned high-carbohydrate (70% kcal carbohydrate) or comparison diet (35% kcal carbohydrate) during six days of short-term moderate-high intensity exercise heat acclimation. HST2 and HST3 were conducted one and five days after completing heat acclimation, respectively. a = different from HST1, $p < 0.05$.

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**SUPPLEMENTAL DIGITAL CONTENT**

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**Figure 1**

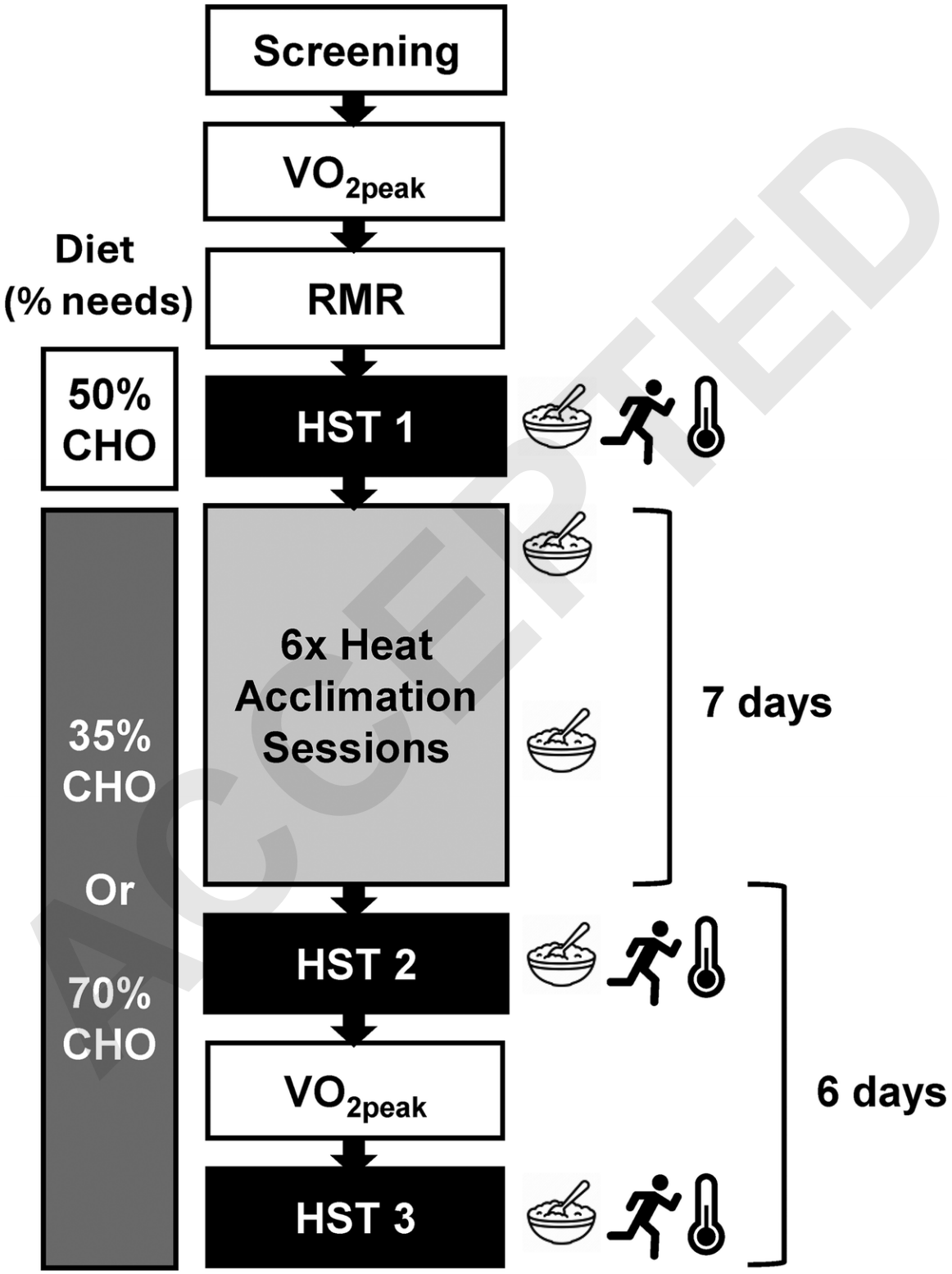


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**Figure 2**

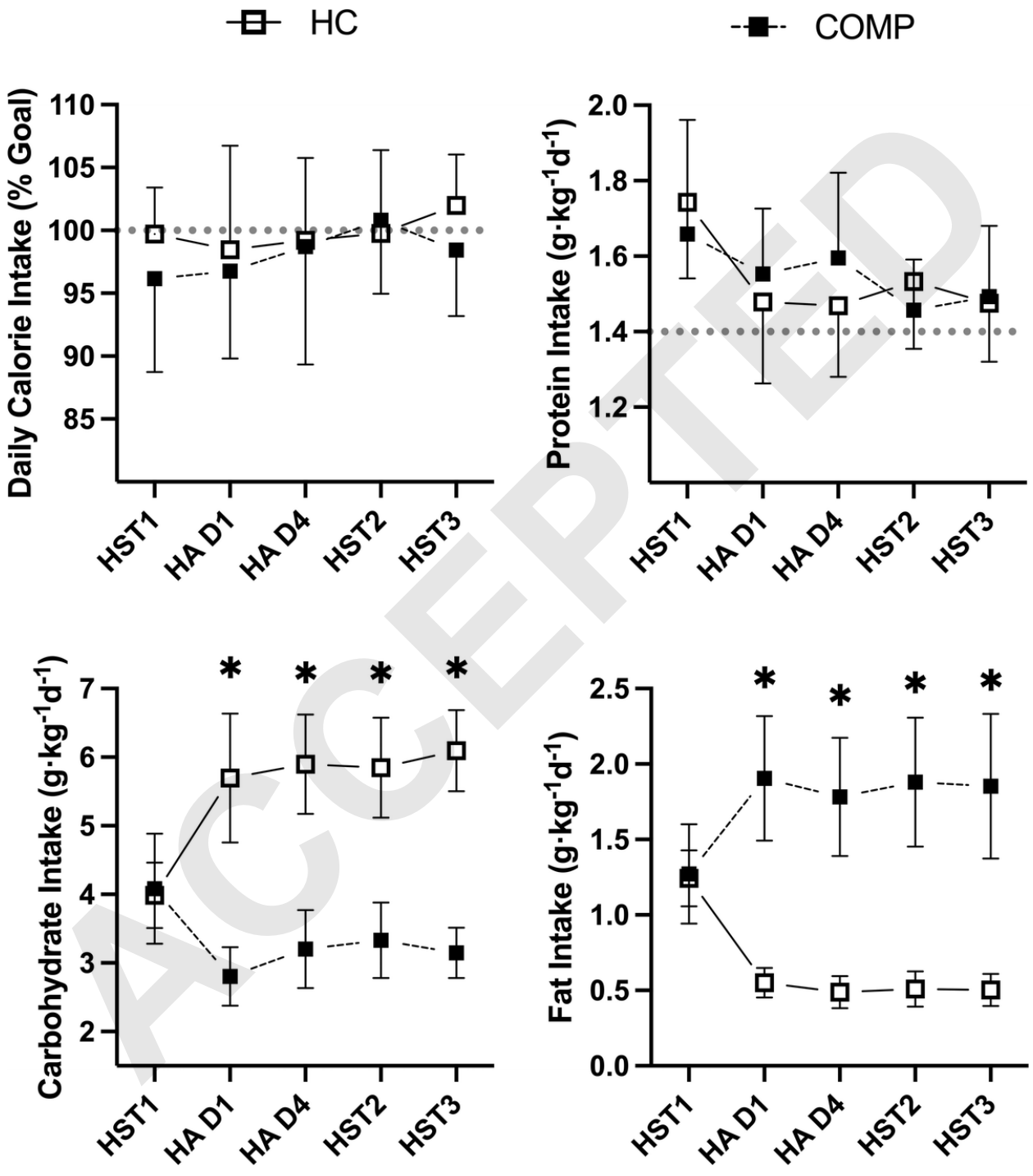


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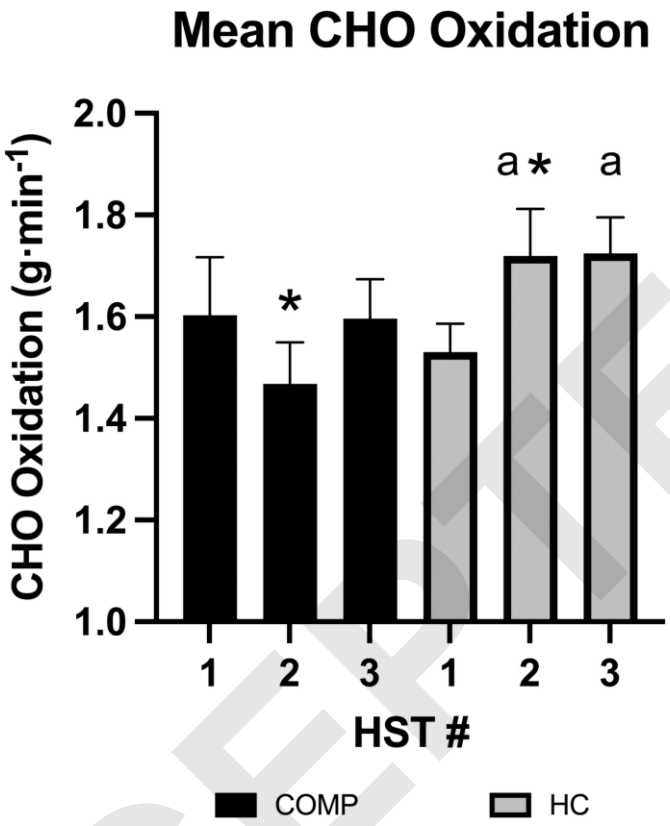
**Figure 3**



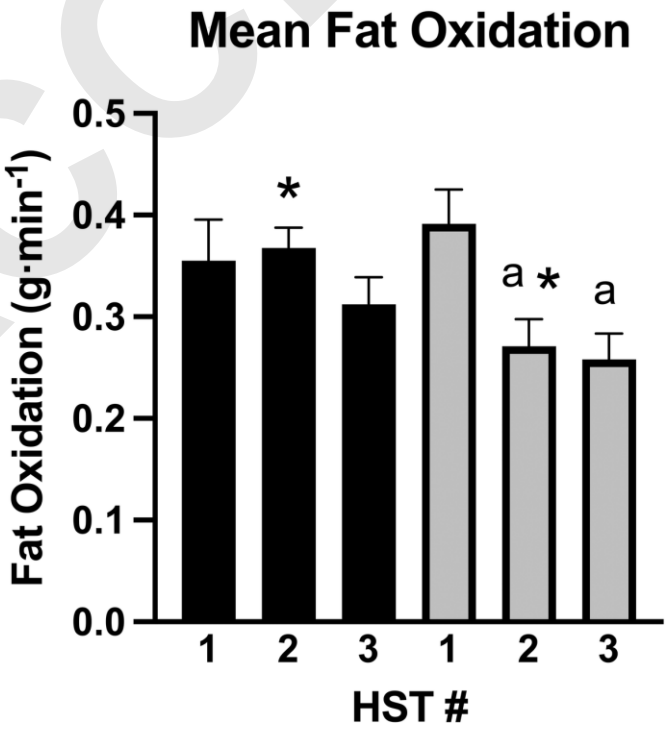




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**Figure 4**

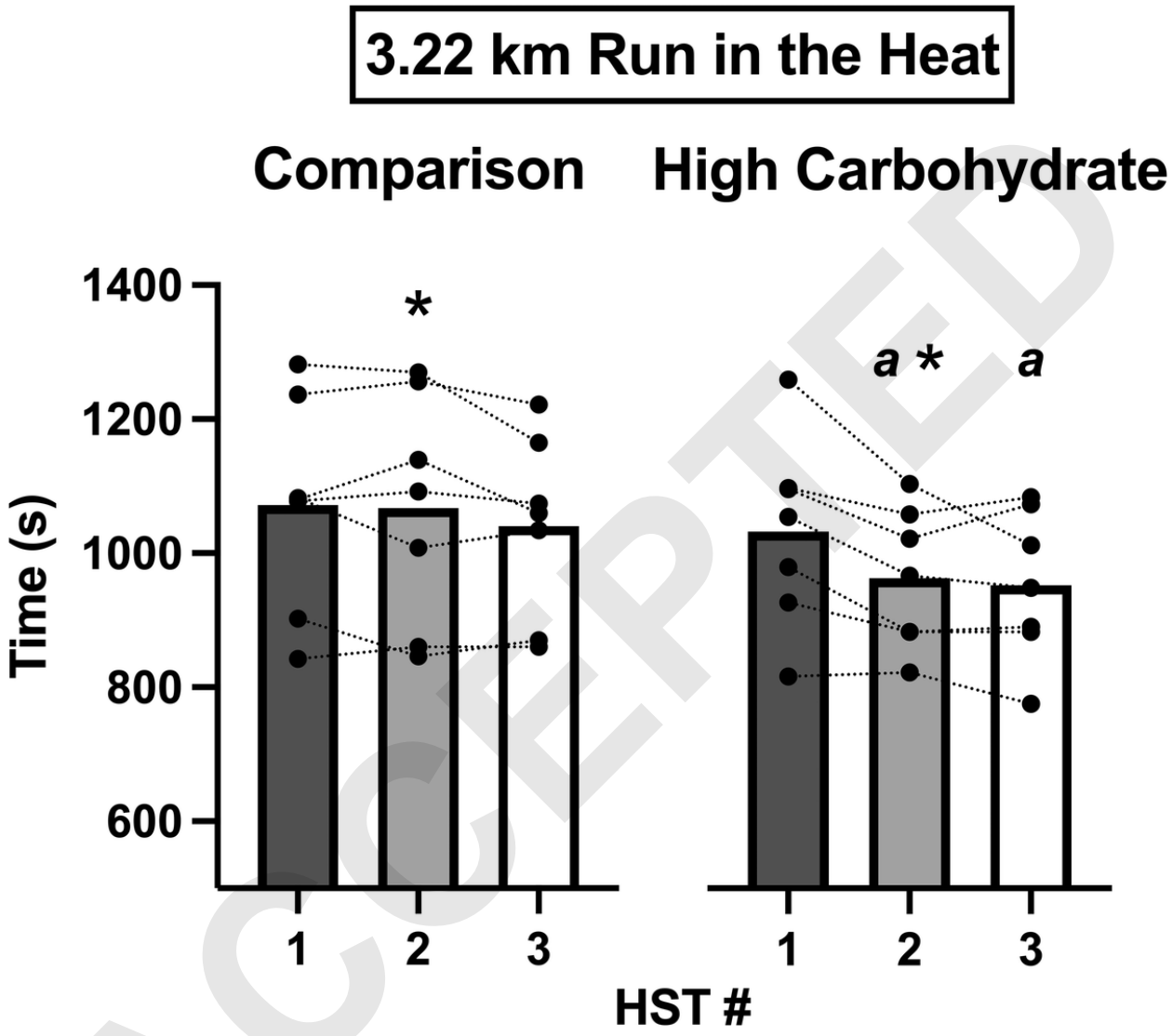


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**Table 1:** Mean nutrition prescription for all participants preceding the initial heat stress test (HST1) and subsequent group interventions during heat acclimation and post-testing

| Variable | Pre-HST1 | High-CHO (HC) | Comparison (COMP) |
|---|---|---|---|
| Energy | RMR x 1.6 to 1.9 activity factor to sustain weight | | |
| Carbohydrate | 4.4 $g \cdot kg^{-1} \cdot day$ | 6.0 $g \cdot kg^{-1} \cdot day$ | 3.1 $g \cdot kg^{-1} \cdot day$ |
| Protein | 1.4 $g \cdot kg^{-1} \cdot day$ | 1.4 $g \cdot kg^{-1} \cdot day$ | 1.4 $g \cdot kg^{-1} \cdot day$ |
| Fat | 1.3 $g \cdot kg^{-1} \cdot day$ | 0.5 $g \cdot kg^{-1} \cdot day$ | 1.9 $g \cdot kg^{-1} \cdot day$ |

NOTE: CHO = Carbohydrate; HST = Heat stress test; RMR = Resting metabolic rate

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**Table 2:** Participant characteristics preceding heat acclimation (HA) with descriptive statistical responses to HA for the high-carbohydrate (HC) intervention group and lower-carbohydrate comparison (COMP) group

| Variable | HC (n = 7) | COMP (n = 7) |
|---|---|---|
| **Participant characteristics preceding HA** | | |
| Age (years) | 26 ± 5 | 25 ± 7 |
| Body mass (kg) | 75.6 ± 6.2 | 72.6 ± 11.7 |
| TBSA ($m^2$) | 1.92 ± 0.09 | 1.85 ± 0.17 |
| Baseline weekly moderate-to-intense training volume (min) | 363 ± 72 | 420 ± 113 |
| Pre-HA $VO_{2peak}$ ($ml \cdot kg^{-1} \cdot min^{-1}$) | 52.7 ± 4.3 | 51.7 ± 4.2 |
| **Descriptive statistics during and in response to HA** | | |
| Daily HA Total Volume (km) | 9.03 ± 0.99 | 8.67 ± 1.28 |
| Daily HA HIIT Run Volume (km) | 3.57 ± 0.51 | 3.76 ± 0.84 |
| Daily HA Time $T_{re} \geq 38.5$ ºC (min) | 67.1 ± 3.5 | 65.7 ± 6.0 |
| Daily HA Heat Exposure (min) | 89.9 ± 7.5 | 92.5 ± 6.8 |
| Post-HA $VO_{2peak}$ ($ml \cdot kg^{-1} \cdot min^{-1}$) | 53.6 ± 4.7 | 51.9 ± 4.9 |

NOTE: CHO = Carbohydrate; HA = Heat acclimation; HIIT = High-intensity interval training; TBSA = Total body surface area; $T_{re}$ = Rectal temperature. There were no between-group differences in any variable ($p > 0.05$).

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**Table 3:** Nutritional and heat stress outcomes for the high-carbohydrate (HC) intervention group and lower-carbohydrate comparison (COMP) at the pre-acclimation heat stress test (HST1) and post-heat acclimation tests (HST2 and HST3).

| Variable | High Carbohydrate Group | | | Comparison Group | | |
|---|---|---|---|---|---|---|
| | HST1 | HST2 | HST3 | HST1 | HST2 | HST3 |
| **Nutritional Comparisons** | | | | | | |
| Body mass (kg) | 75.4 ± 6.5 | 74.9 ± 6.6 | 75.0 ± 6.6 | 72.5 ± 12.3 | 72.2 ± 12.0 | 72.3 ± 11.8 |
| Carbohydrate ($g \cdot kg^{-1} d^{-1}$) | 4.0 ± 0.5 | 5.8 ± 0.8 [a]* | 6.1 ± 0.6 [a]* | 4.1 ± 0.8 | 3.3 ± 0.6 [a]* | 3.2 ± 0.4 [a]* |
| Protein ($g \cdot kg^{-1} d^{-1}$) | 1.7 ± 0.2 | 1.5 ± 0.2 | 1.5 ± 0.2 | 1.7 ± 0.3 | 1.5 ± 0.1 | 1.5 ± 0.2 |
| Fat ($g \cdot kg^{-1} d^{-1}$) | 1.2 ± 0.2 | 0.5 ± 0.1 [a]* | 0.5 ± 0.1 [a]* | 1.3 ± 0.1 | 1.9 ± 0.4 [a]* | 1.9 ± 0.5 [a]* |
| Energy (% goal) | 99.7 ± 4.0 | 99.8 ± 7.2 | 102.0 ± 4.4 | 96.2 ± 8.0 | 100.8 ± 6.3 | 98.4 ± 5.7 |
| **Indices of Heat Acclimation** | | | | | | |
| End-Walk HR (bpm) | 153 ± 8 | 139 ± 10 [a] | 138 ± 10 [a] | 158 ± 11 | 145 ± 8 [a] | 145 ± 7 [a] |
| End-Walk $T_{re}$ (ºC) | 38.35 ± 0.29 | 38.00 ± 0.20 [a] | 38.09 ± 0.30 [a] | 38.44 ± 0.34 | 38.07 ± 0.42 [a] | 38.21 ± 0.40 [a] |
| End-Walk $T_{sk}$ (ºC) | 36.69 ± 0.54 | 36.09 ± 0.55 [a] | 36.15 ± 0.57 [a] | 36.83 ± 0.47 | 36.01 ± 0.69 [a] | 36.28 ± 0.52 [a] |
| End-Walk $\dot{V}O_2$ ($ml \cdot kg^{-1} \cdot min^{-1}$) | 26.3 ± 3.4 | 24.2 ± 3.3 [a] | 24.1 ± 3.5 [a] | 26.6 ± 4.2 | 25.3 ± 4.0 [a] | 25.3 ± 4.0 |
| End-Walk Lactate ($mmol \cdot L^{-1}$) | 2.5 ± 1.1 | 1.2 ± 0.5 [a] | 1.6 ± 0.5 | 3.2 ± 1.6 | 1.6 ± 0.7 [a] | 2.1 ± 0.9 |
| Sweat rate ($L \cdot hr^{-1}$) | 1.07 ± 0.29 | 1.18 ± 0.22 [a] | 1.17 ± 0.31 [a] | 1.15 ± 0.27 | 1.25 ± 0.26 [a] | 1.25 ± 0.21 [a] |

NOTE: Physiological and perceptual values represent the end of the 45-minute walk. Sweat rate was recorded after completion of all testing. HR = Heart rate; $T_{re}$ = Rectal temperature; $T_{sk}$ = Mean skin temperature. a = different from HST1; ***** = different between groups, $p < 0.05$

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**Table 4:** Running performance outcomes between high-carbohydrate (HC) intervention group and lower-carbohydrate comparison (COMP) at the pre-acclimation heat stress test (HST1) and post-heat acclimation tests (HST2 and HST3).

| Variable | High Carbohydrate Group | | | Comparison Group | | |
|---|---|---|---|---|---|---|
| | HST1 | HST2 | HST3 | HST1 | HST2 | HST3 |
| **Run Performance Metrics** | | | | | | |
| 3.22 km TT run time (s) | 1029 ± 141 | 959 ± 103 [a] * | 949 ± 111 [a] | 1072 ± 160 | 1067 ± 172 * | 1041 ± 136 |
| 0 to 1.61 km run time (s) | 519 ± 75 | 496 ± 60 | 486 ± 59 | 523 ± 65 | 532 ± 82 | 505 ± 57 |
| 1.61 to 3.22 km run time (s) | 509 ± 76 | 466 ± 48 [a] * | 463 ± 55 [a] * | 549 ± 100 | 534 ± 102 * | 537 ± 89 * |
| Change in run time from HST1 (s) | | - 69 ± 51 * | - 80 ± 80 | | - 5 ± 47 * | - 31 ± 43 |
| **Running Strain Indices** | | | | | | |
| End-Run HR (bpm) | 194 ± 4 | 192 ± 5 | 191 ± 6 | 193 ± 8 | 188 ± 9 [a] | 190 ± 6 |
| Mean Run % $HR_{max}$ | 99.9 ± 4.1 | 97.9 ± 4.8 | 98.0 ± 3.8 | 100.2 ± 4.0 | 96.4 ± 4.3 | 97.7 ± 4.3 |
| End-Run $T_{re}$ (ºC) | 39.21 ± 0.28 | 39.04 ± 0.38 | 39.10 ± 0.42 | 39.24 ± 0.32 | 39.12 ± 0.19 | 39.08 ± 0.35 |
| End-Run RPE (6-20, AU) | 18 ± 2 | 18 ± 1 | 18 ± 1 | 19 ± 1 | 18 ± 1 | 19 ± 1 |

NOTE: HR = Heart rate; $T_{re}$ = Rectal temperature; TT = Time trial. Mean run %$HR_{max}$ was calculated from 25-100% of run duration. a = different from HST1, * = different between groups, $p < 0.05$

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**Supplementary Table:** Absolute nutritional intake for the high-carbohydrate (HC) intervention group and lower-carbohydrate comparison (COMP) at the pre-acclimation heat stress test (HST1) and post-heat acclimation tests (HST2 and HST3).

| Variable | High Carbohydrate Group | | | | Comparison Group | | | |
|---|---|---|---|---|---|---|---|---|
| | HST1 | HA WEEK | HST2 | HST3 | HST1 | HA WEEK | HST2 | HST3 |
| Carbohydrate (g) | 300 ± 42 | 436 ± 56 [a]* | 440 ± 70 [a]* | 458 ± 52 [a]* | 292 ± 29 | 216 ± 20 [a]* | 239 ± 24 [a]* | 227 ± 19 [a]* |
| Protein (g) | 131 ± 20 | 112 ± 21 | 116 ± 18 | 111 ± 19 | 120 ± 22 | 114 ± 17 | 107 ± 22 | 108 ± 17 |
| Fat (g) | 93 ± 13 | 39 ± 8 [a]* | 38 ± 9 [a]* | 38 ± 9 [a]* | 91 ± 23 | 132 ± 14 [a]* | 134 ± 17 [a]* | 131 ± 18 [a]* |
| Energy (kcal) | 2563 ± 268 | 2545 ± 317 | 2567 ± 340 | 2523 ± 163 | 2467 ± 239 | 2504 ± 193 | 2586 ± 220 | 2620 ± 265 |
| HA Session CHO (g) | -- | 55 ± 12 | -- | -- | -- | 48 ± 16 | -- | -- |

NOTE: HA = Heat acclimation. a = different from HST1; ***** = different between groups, $p < 0.05$. HA Session CHO reflects the mean amount of CHO consumed from CHO-electrolyte beverages during HA sessions.

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